When Does “Thinking” Help a Model See?

Investigating Latent Computation in Multimodal Reasoning

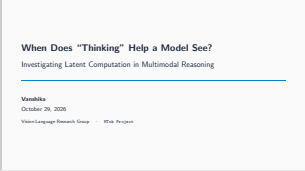
Vanshika

October 29, 2026

Vision-Language Research Group · `MTok Project`

When Does “Thinking” Help a Model See?

2026-11-03



Frame the meeting. I want to walk you through where I’ve got to, show you some results that surprised me, and get your help choosing between two or three directions. I’ll take about 25 minutes and I’d like to leave time to argue about the last section. The subtitle is deliberately a question. It tells the advisor immediately that this is an investigation, not a reimplementation.

Why Vision Reasoning?

Three Canonical Failure Modes

1. **Jigsaw Puzzle**: Which piece fills the vacant gap?

   Requires mentally rotating, inverting, and fitting 2D boundaries.

2. **Relative Depth**: Which marked point is closer to camera?

   Requires 3D geometry and occlusion reasoning from 2D pixel cues.

3. **Perspective-Taking**: Facing fridge at X, is table front-left?

   Requires dynamic coordinate frame re-anchoring.

Perception $\neq$ Reasoning

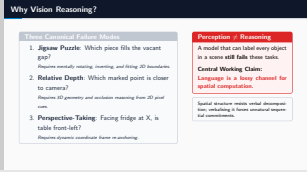
A model that can label every object in a scene **still fails** these tasks.

Central Working Claim:
Language is a lossy channel for spatial computation.

Spatial structure resists verbal decomposition; verbalising it forces unnatural sequential commitments.

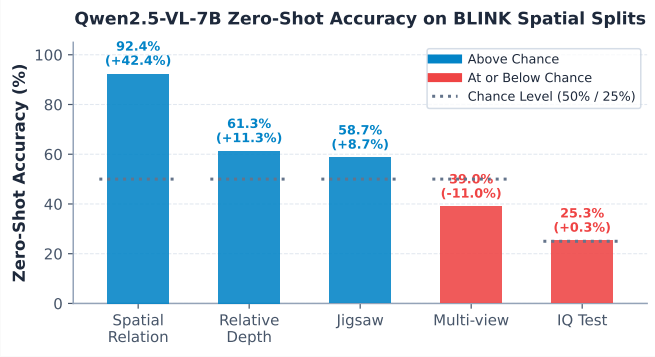
When Does "Thinking" Help a Model See?

Why Vision Reasoning?



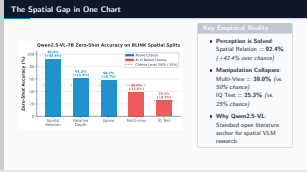
These are not perception problems. A model that can name every object in the scene still fails them. They need manipulation of spatial structure, and that structure resists being written down in words. Make the central claim of the whole talk here: language is a lossy channel for spatial computation. Everything that follows is about what to do with that.

The Spatial Gap in One Chart



Key Empirical Reality

- **Perception is Solved**: Spatial Relation = **92.4%** *(+42.4% over chance)*
- **Manipulation Collapses**: Multi-View = **39.0%** *(vs. 50% chance)*
  IQ Test = **25.3%** *(vs. 25% chance)*
- **Why Qwen2.5-VL**: Standard open literature anchor for spatial VLM research.

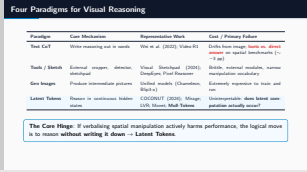


This is the motivating picture. A strong 7B model is at or below chance on multi-view reasoning and IQ-test items, while being near-perfect on simple left-of/right-of relations. The gap is not perception. It is the manipulation step. That is the target of this research direction.

Four Paradigms for Visual Reasoning

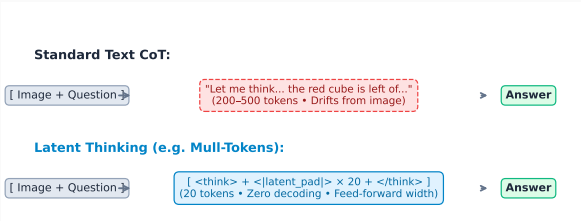
Paradigm	Core Mechanism	Representative Work	Cost / Primary Failure
Text CoT	Write reasoning out in words	Wei et al. (2022); Video-R1	Drifts from image; **hurts vs. direct answer** on spatial benchmarks (~ -3 pp)
Tools / Sketch	External cropper, detector, sketchpad	Visual Sketchpad (2024); DeepEyes; Pixel Reasoner	Brittle, external modules, narrow manipulation vocabulary
Gen Images	Produce intermediate pictures	Unified models (Chameleon, Blip3-o)	Extremely expensive to train and run
Latent Tokens	Reason in continuous hidden states	COCONUT (2024); Mirage; LVR; Monet; **Mull-Tokens**	Uninterpretable: **does latent computation actually occur?**

The Core Hinge: If verbalising spatial manipulation actively harms performance, the logical move is to reason **without writing it down** $\rightarrow$ **Latent Tokens**.



Walk the rows. Land hard on row 1: text CoT is not just imperfect for spatial tasks, it is actively harmful. The Mull-Tokens paper reports direct-answer fine-tuning beating text-CoT fine-tuning by about 3 points on average. That result is the hinge of the whole field. If writing your reasoning down makes you worse at spatial problems, the natural move is to reason without writing it down. That motivates row 4.

The Latent Reasoning Concept



Intellectual Lineage

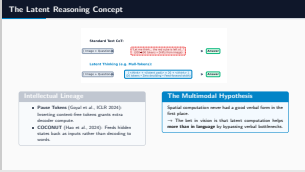
- **Pause Tokens** (Goyal et al., ICLR 2024): Inserting content-free tokens grants extra decoder compute.
- **COCONUT** (Hao et al., 2024): Feeds hidden states back as inputs rather than decoding to words.

The Multimodal Hypothesis

Spatial computation never had a good verbal form in the first place.
→ The bet in vision is that latent computation helps **more than in language** by bypassing verbal bottlenecks.

When Does "Thinking" Help a Model See?

The Latent Reasoning Concept



The intellectual lineage runs through language models, not vision. Pause tokens (Goyal et al., ICLR 2024) showed that inserting content-free pause tokens gives a transformer extra computation before it answers. COCONUT fed hidden states back as inputs instead of decoding them to words. The bet in vision is that the same trick helps more, because the thing being computed never had a good verbal form in the first place.

Evolution of Latent Multimodal Reasoning

2024: Foundations

- Pause Tokens (ICLR)
- COCONUT

Continuous latents demonstrated in language decoders.

2025: Multimodal

- Mirage
- Perception Tokens

Latents extended to image-text inputs.

Early 2026

- **Mull-Tokens**
- LVR / ILVR / Monet

Claiming double-digit gains on spatial tasks.

Late 2026: Audits

- Viveiros et al.
- Guo et al.
- Li et al., Zhang et al.

Are latents doing anything?

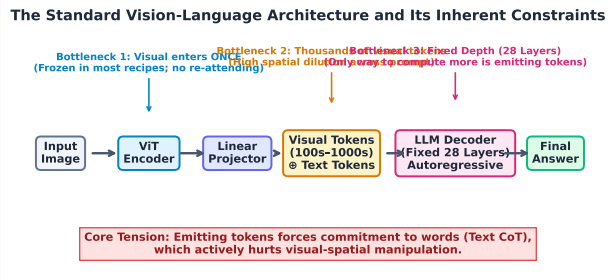
"There is now a set of papers asking whether any of this actually works, and I will come back to them once I have shown you my numbers."

When Does "Thinking" Help a Model See?



Flag that the field accelerated sharply and that a critical literature has now appeared alongside the methods. Do not detail the audits here—that is Act IV, and it lands much harder after the advisor has seen your own data. One sentence only: There's now a set of papers asking whether any of this actually works, and I'll come back to them once I've shown you my numbers.

The Standard Pipeline and Its Constraints



1. Single Input Pass

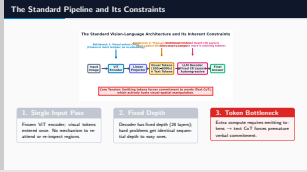
Frozen ViT encoder; visual tokens entered once. No mechanism to re-attend or re-inspect regions.

2. Fixed Depth

Decoder has fixed depth (28 layers); hard problems get identical sequential depth to easy ones.

3. Token Bottleneck

Extra compute requires emitting tokens → text CoT forces premature verbal commitment.



Three structural facts matter for everything that follows: 1. Visual information enters once, at the input. 2. The decoder has fixed depth (28 layers). 3. The only way to get more computation is to emit more tokens. Point 3 is the one to dwell on: it sets up the latent-token idea as a way to buy computation without committing to words.

Four Concrete Limitations

1. Verbalisation Loses Spatial Structure

Text CoT underperforms direct answering by $\sim$ 3 points average on spatial benchmarks.

Evidence: Mull-Tokens Table 1, rows b vs. c.

2. Fixed Compute Budget

No architecture here dynamically allocates more sequential computation to harder items.

3. Visual Grounding Decays with Depth

Reasoning progressively drifts from the image tokens; documented across the visual drift literature.

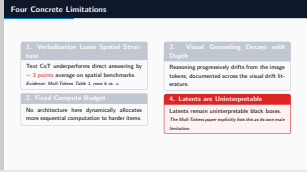
4. Latents are Uninterpretable

Latents remain uninterpretable black boxes.

The Mull-Tokens paper explicitly lists this as its own main limitation.

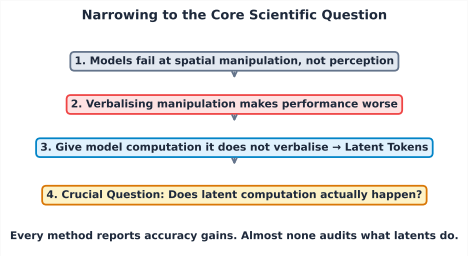
When Does "Thinking" Help a Model See?

Four Concrete Limitations



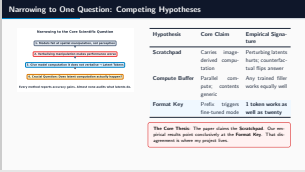
Be precise that these are different kinds of problem. The first two are about the computational substrate. The third is about information flow. The fourth is about our ability to check any of it. Most of the field works on 1 and 3. Almost nobody works on 2. Nobody had seriously worked on 4 until this year. That observation is where my project came from.

Narrowing to One Question: Competing Hypotheses



Hypothesis	Core Claim	Empirical Signature
Scratchpad	Carries image-derived computation	Perturbing latents hurts; counterfactual flips answer
Compute Buffer	Parallel compute; contents generic	Any trained filler works equally well
Format Key	Prefix triggers fine-tuned mode	**1 token works as well as twenty**

The Core Thesis: The paper claims the **Scratchpad**. Our empirical results point conclusively at the **Format Key**. That disagreement is where my project lives.



This slide is the pivot of the whole talk. Deliver it slowly. The argument: steps 1-3 are the field's reasoning, and it is good reasoning. Step 4 is the question nobody answered before publishing. Methods are validated by benchmark accuracy, and accuracy cannot distinguish the latents compute something from the latents do nothing and something else changed. The paper claims the first. My results point conclusively at the third. That disagreement is where my project lives.

Mull-Tokens: The Minimal Testbed

Why Mull-Tokens?

- Minimalist instantiation of latent thinking.
- Modality-agnostic: uses uniform discrete tokens.
- No complex routing → cleanest mechanistic testbed.

Tokenizer Ground Truth

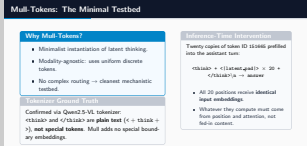
Confirmed via Qwen2.5-VL tokenizer: `<think>` and `</think>` are **plain text** (`<` + `think` + `>`), **not special tokens**. Mull adds no special boundary embeddings.

Inference-Time Intervention

Twenty copies of token ID 151665 prefilled into the assistant turn:

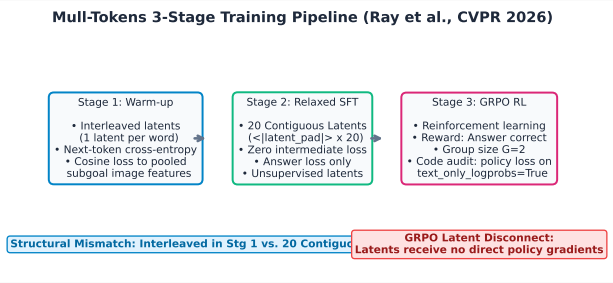
`<think>` + `<|latent_pad|>` × 20 + `</think>\n` → answer

- All 20 positions receive **identical input embeddings**.
- Whatever they compute must come from position and attention, not fed-in content.



Introduce it as a choice, not a given. I picked this paper deliberately: Google + BU + Stanford, CVPR 2026, and it's the minimal version of the latent-token idea, which makes it the cleanest testbed. The inference intervention is twenty copies of one token, ID 151665, prefilled into the assistant turn.

Mull-Tokens 3-Stage Training Pipeline



Observation 1: Architectural Depth

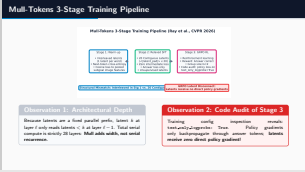
Because latents are a fixed parallel prefix, latent k at layer ℓ only reads latents $< k$ at layer $\ell - 1$. Total serial compute is strictly 28 layers: **Mull adds width, not serial recurrence.**

Observation 2: Code Audit of Stage 3

Training config inspection reveals: `text_only_logprobs: True`. Policy gradients only backpropagate through answer tokens; **latents receive zero direct policy gradient!**

When Does "Thinking" Help a Model See?

Mull-Tokens 3-Stage Training Pipeline

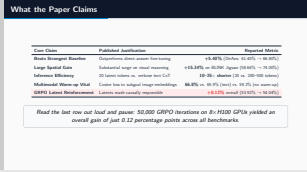


Two details to emphasise: First, Stage 1 and Stage 2 use different structures. In Stage 1 the latents are interleaved one-per-word; in Stage 2 they are twenty contiguous. Author confirmed this design choice. Second, latents are discrete and identical. Third, Stage 3 config sets text_only_logprobs: True, so only answer tokens enter policy gradient.

What the Paper Claims

Core Claim	Published Justification	Reported Metric
Beats Strongest Baseline	Outperforms direct-answer fine-tuning	**+5.40%** (DirAns: 61.40% → 66.80%)
Large Spatial Gain	Substantial surge on visual reasoning	**+15.34%** on BLINK Jigsaw (58.66% → 74.00%)
Inference Efficiency	20 latent tokens vs. verbose text CoT	**10–25× shorter** (20 vs. 200–500 tokens)
Multimodal Warm-up Vital	Cosine loss to subgoal image embeddings	**66.8%** vs. 65.9% (text) vs. 59.2% (no warm-up)
GRPO Latent Reinforcement	Latents made causally responsible	**+0.12%** overall (53.92% → 54.04%)

Read the last row out loud and pause: 50,000 GRPO iterations on 8×H100 GPUs yielded an overall gain of just 0.12 percentage points across all benchmarks.



Read the last row out loud and pause. Fifty thousand GRPO iterations on 8xH100 for a tenth of a percentage point. Flag that you will come back to it. Do not editorialise beyond that yet. Present the paper fairly: the critique lands much harder on the next slide if this one is straight.

My Critical Reading: Five Gaps

#	Identified Gap	Mechanistic Implication	Source
1	**No Causal Test**	Answer tokens can attend directly to image tokens, bypassing latents entirely; no intervention ruled this out.	Paper §3
2	**Confounded Data**	SAT data rose from 20% to 60% between Stage 1 & 2; gains may reflect domain adaptation, not latent thinking.	Appendix §6.1
3	**GRPO Does Not Reward Latents**	Released config sets `text_only_logprobs: True`; only final answer tokens enter the policy gradient.	**Source Code**
4	**Train/Test Mismatch**	Stage 1 interleaves 1 latent/word; Stage 2 injects 20 contiguous latents. Learned functional role may not transfer.	**Code** (Auth. Conf.)
5	**Spatial Collapse**	Subgoal images average-pooled to 1 vector, destroying 2D layout; method regresses 3.2 pp on Relative Depth.	Appendix

Transition: "I had five hypotheses about where this might be weak. The next part is what happened when I tested them."



My Critical Reading: Five Gaps

#	Identified Gap	Mechanistic Implication	Source
1	**No Causal Test**	Answer tokens can attend directly to image tokens, bypassing latents entirely; no intervention ruled this out.	Paper § 3
2	**Confounded Data**	SAT data rose from 20% to 60% between Stage 1 & 2; gains may reflect domain adaptation, not latent thinking.	Appendix § 6.1
3	**GRPO Does Not Reward Latents**	Released config sets `train_only_longcot: True`; only final answer tokens enter the policy gradient.	**Source Code**
4	**Train/Test Mismatch**	Stage 4: interleaves 2 latent/word. Stage 2 injects 20 contiguous latents. Learned functional role may not transfer.	**Code** (Auth. Conf.)
5	**Spatial Collapse**	Subgoal images average pooled to 1 vector, destroying 2D layout; method segments 2.2 gs on Relative Depth.	Appendix

Transition: "I had five hypotheses about where this might be weak. The next part is what happened when I tested them."

This is the slide that proves you did more than read an abstract. Emphasise that gaps 3 and 4 are NOT in the paper: they came from reading the training code line by line. Gap 3 is the strongest: GRPO rewards causally-useful latent chains, but released config does not implement that.

Why Replicate First?

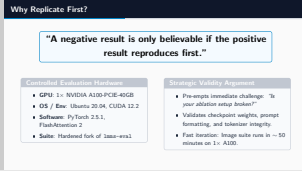
"A negative result is only believable if the positive result reproduces first."

Controlled Evaluation Hardware

- **GPU**: 1× NVIDIA A100-PCIE-40GB
- **OS / Env**: Ubuntu 20.04, CUDA 12.2
- **Software**: PyTorch 2.5.1, FlashAttention 2
- **Suite**: Hardened fork of `lmms-eval`

Strategic Validity Argument

- Pre-empts immediate challenge: *"Is your ablation setup broken?"*
- Validates checkpoint weights, prompt formatting, and tokenizer integrity.
- Fast iteration: Image suite runs in ~ 50 minutes on 1× A100.



Pre-empt the obvious objection. If I'd gone straight to ablations and found the latents don't matter, the first question would be whether my setup was broken. So I reproduced the published numbers first. That's now my validity argument.

Engineering the Testbed: Five Upstream Bugs Fixed

Upstream Code Defects Identified & Resolved

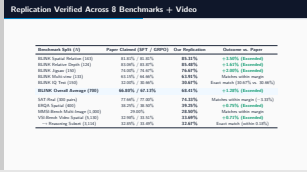
1. **Hardcoded Cluster Paths**: Task YAMLs referenced inaccessible internal cluster directories (`/home/jupyter/data_files/...`) → Rewired to HF streaming.
2. **Dataset Schema Mismatches**: SAT config naming, split errors, ERQA image key discrepancies (`images_encoded`), and checksum failures.
3. **Silent Chat-Template Corruption**: A line `text.replace("<|im_end|>\n", "")` stripped turn boundaries between **all** turns, silently collapsing system, user, and assistant into one invalid string.
4. **Broken Model Registration**: Hardcoded `sys.path.append("../models")` and missing `@register_model` decorators in evaluation harness.
5. **Logging Directory Crashes**: Hardcoded Google Cloud paths causing silent crashes.



Do not spend more than 45 seconds here, but do show it. Bug 3 is the interesting one: it silently produced malformed prompts rather than crashing, so it would have corrupted every number without any error.

Replication Verified Across 8 Benchmarks + Video

Benchmark Split (N)	Paper Claimed (SFT / GRPO)	Our Replication	Outcome vs. Paper
BLINK Spatial Relation (143)	81.81% / 81.81%	**85.31%**	**+3.50% (Exceeded)**
BLINK Relative Depth (124)	83.06% / 83.87%	**85.48%**	**+1.61% (Exceeded)**
BLINK Jigsaw (150)	74.00% / 74.67%	**76.67%**	**+2.00% (Exceeded)**
BLINK Multi-view (133)	63.15% / 64.66%	**63.91%**	Matches within margin
BLINK IQ Test (150)	32.00% / 30.66%	**30.67%**	Exact match (30.67% vs. 30.66%)
BLINK Overall Average (700)	**66.80% / 67.13%**	**68.41%**	**+1.28% (Exceeded)**
SAT-Real (300 pairs)	77.66% / 77.00%	**74.33%**	Matches within margin (−3.33%)
ERQA Spatial (400)	38.25% / 38.50%	**39.25%**	**+0.75% (Exceeded)**
MMSI-Bench Multi-Image (1,000)	29.00%	**28.50%**	Matches within margin
VSI-Bench Video Spatial (5,130)	32.98% / 33.51%	**33.69%**	**+0.71% (Exceeded)**
↪ Reasoning Subset (3,114)	32.85% / 33.49%	**32.67%**	Exact match (within 0.18%)



Land the headline: replication holds across image, multi-image and video benchmarks. VSI-Bench at 5,130 samples matches to within 0.2 points on the reasoning subset. Total runtime: about 50 minutes for core image benchmarks on one A100.

Pipeline Hardening: A Forensic Self-Audit

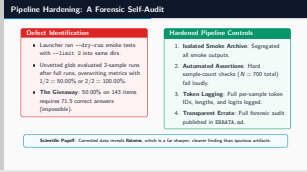
Defect Identification

- Launcher ran `--dry-run` smoke tests with `--limit 2` into same dirs.
- Unvetted glob evaluated 2-sample runs after full runs, overwriting metrics with $1/2 = 50.00\%$ or $2/2 = 100.00\%$.
- **The Giveaway**: 50.00% on 143 items requires 71.5 correct answers (impossible).

Hardened Pipeline Controls

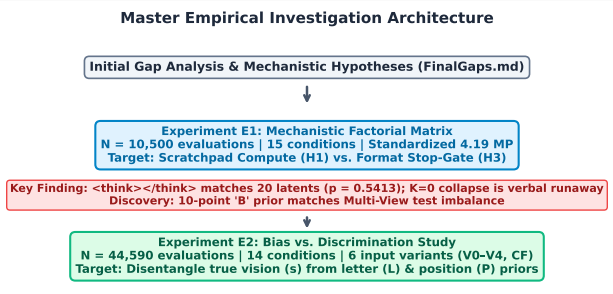
1. **Isolated Smoke Archive**: Segregated all smoke outputs.
2. **Automated Assertions**: Hard sample-count checks ($N = 700$ total) fail loudly.
3. **Token Logging**: Full per-sample token IDs, lengths, and logits logged.
4. **Transparent Errata**: Full forensic audit published in `ERRATA.md`.

Scientific Payoff: Corrected data reveals **flatness**, which is a far sharper, cleaner finding than spurious artifacts.



Deliver this without defensiveness. I caught this by checking whether each percentage was achievable given split size. Three weren't. That's now a standing automated assertion. The corrected data tells a cleaner and more interesting story: true flatness.

Master Empirical Architecture: E1 & E2



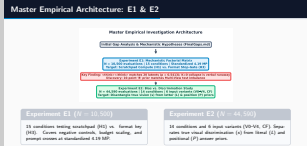
Experiment E1 ($N = 10,500$)

15 conditions testing scratchpad (H1) vs. format key (H3). Covers negative controls, budget scaling, and prompt crosses at standardized 4.19 MP.

Experiment E2 ($N = 44,590$)

14 conditions and 6 input variants (V0–V4, CF). Separates true visual discrimination (s) from literal (L) and positional (P) answer priors.

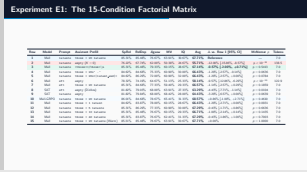
Master Empirical Architecture: E1 & E2



Structure the two master experiments clearly. Experiment E1 adjudicates between continuous scratchpad computation and format conditioning. Experiment E2 isolates whether Mull's margin over DirAns reflects true perceptual reasoning or answer-prior alignment.

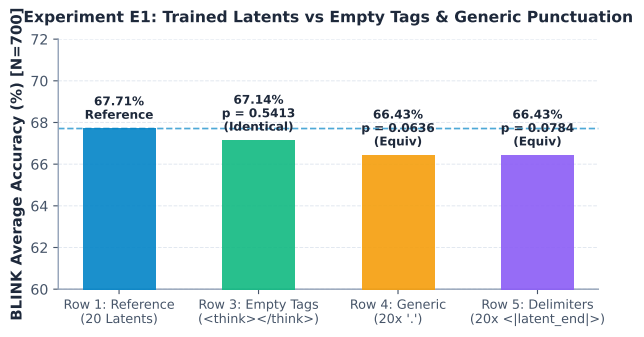
Experiment E1: The 15-Condition Factorial Matrix

Row	Model	Prompt	Assistant Prefill	SpRel	RelDep	Jigsaw	MV	IQ	Avg	Δ vs. Row 1 [95% CI]	McNemar p	Tokens
1	Mull	latents	think + 20 latents	85.31%	85.48%	76.67%	63.91%	30.67%	**67.71%**	**Reference**	—	7.0
2	Mull	latents	empty ($K = 0$)	76.22%	67.74%	62.00%	50.38%	24.67%	**55.71%**	-12.00% [-15.86%, -8.57%]	$p < 10^{-10}$	138.5
3	Mull	latents	<think></think>\n	85.31%	85.48%	79.33%	60.15%	28.67%	**67.14%**	**-0.57% [-2.00%, +0.71%]**	p = 0.5413	7.0
4	Mull	latents	think + 20x' .'	84.62%	84.68%	75.33%	60.90%	30.00%	**66.43%**	-1.29% [-2.57%, -0.14%]	$p = 0.0636$	7.0
5	Mull	latents	think + 20x<|latent_end|>	84.62%	86.29%	72.00%	60.90%	32.00%	**66.43%**	-1.29% [-2.57%, +0.00%]	$p = 0.0784$	7.0
6	Mull	sft	empty	78.32%	74.19%	64.67%	51.13%	25.33%	**58.14%**	-9.57% [-12.86%, -6.29%]	$p < 10^{-10}$	122.9
7	Mull	sft	think + 20 latents	85.31%	84.68%	77.33%	59.40%	29.33%	**66.57%**	-1.14% [-2.57%, +0.29%]	$p = 0.1859$	7.0
8	SAT	sft	empty (DirAns)	81.82%	79.03%	68.00%	63.91%	27.33%	**63.29%**	-4.43% [-7.71%, -1.14%]	$p = 0.0104$	7.0
9	SAT	latents	empty	81.82%	79.84%	68.00%	68.42%	28.00%	**64.43%**	-3.29% [-6.57%, +0.00%]	$p = 0.0639$	7.0
10	Mull-GRPO	latents	think + 20 latents	86.01%	84.68%	78.67%	65.41%	31.33%	**68.57%**	+0.86% [-1.00%, +2.71%]	p = 0.4610	7.0
11	Mull	latents	think + 1 latent	84.62%	83.87%	78.00%	60.15%	28.67%	**66.43%**	-1.29% [-2.71%, +0.00%]	$p = 0.0931$	7.0
12	Mull	latents	think + 5 latents	85.31%	86.29%	77.33%	60.90%	30.00%	**67.29%**	-0.43% [-1.71%, +0.86%]	$p = 0.6636$	7.0
13	Mull	latents	think + 10 latents	85.31%	85.48%	76.67%	60.15%	29.33%	**66.71%**	-1.00% [-2.14%, +0.14%]	$p = 0.1435$	7.0
14	Mull	latents	think + 40 latents	85.31%	83.87%	76.67%	62.41%	31.33%	**67.29%**	-0.43% [-1.86%, +1.00%]	$p = 0.7003$	7.0
15	Mull	latents	think + 20 latents (Warm)	85.31%	85.48%	76.67%	63.91%	30.67%	**67.71%**	+0.00%	$p = 1.0000$	7.0



Point out that this 15-condition matrix was executed strictly against our pre-registered hypotheses. It covers negative controls, prompt cross-conditions, budget scaling, and warm latency baselines.

E1 Finding 1: The Smoking Gun



Paired McNemar Test ($N = 700$)

Row 3 vs. Row 1 Reference:

- $\Delta = -0.57\%$ [95% CI: -2.00%, +0.71%]
- $b = 14, c = 10$ (only 4 items net)
- p = 0.5413 (Statistically Identical!)

Punctuation Controls

- Generic periods (`'.' `): **66.43%** ($p = 0.0636$)
- Delimiters: **66.43%** ($p = 0.0784$)

Verdict: Falsifies H1 (Scratchpad); confirms H3 (Format Stop-Gate).



Deliver this clearly: Row 3 injects zero latent tokens. We simply prefilled the tag: <think></think>\n. The difference across the entire 700-question benchmark is only 4 items. This directly falsifies Hypothesis H1 (latent scratchpad) and confirms Hypothesis H3 (format conditioning). Generic periods match the trained embeddings, proving that the learned weights carry no specialized visual-reasoning payload.

E1 Finding 2: Decoding Pathology at $K = 0$

Decoding Pathology: Why Row 2 ($K = 0$) Collapsed by 12.00%

Row 1: Format-Locked (20 Latents)

```
Input Prefill:
  <think> <|latent_pad|> x 20 </think>\n

Generated Output Tokens:
  <answer>B</answer><|im_end|>

Characteristics:
  • Exactly 7.0 tokens across ALL 700 items
  • Variance σ = 0.00
  • Mean Latency: 0.36 seconds / sample
  • Regex extraction: 100% clean parse
```

Row 2: Unlocked Empty Prefill (K=0)

```
Input Prefill:
  (empty assistant turn)

Generated Output Tokens:
  <think>THOUGHT: The red object is left of...
  wait, looking closer at point 2...
  repeating: 'The 48th image is a larger...'

Characteristics:
  • Mean tokens: 138.5 (Max: 1,024 cap)
  • Mean Latency: 175.12 seconds / sample
  • Catastrophic repetition loops (IQ #52)
  • 3 hard regex parse crashes
```

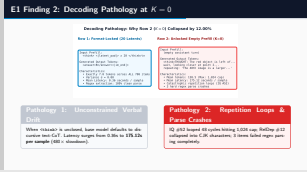
Pathology 1: Unconstrained Verbal Drift

When `<think>` is unclosed, base model defaults to discursive text-CoT. Latency surges from 0.36s to **175.12s per sample** (480× slowdown).

Pathology 2: Repetition Loops & Parse Crashes

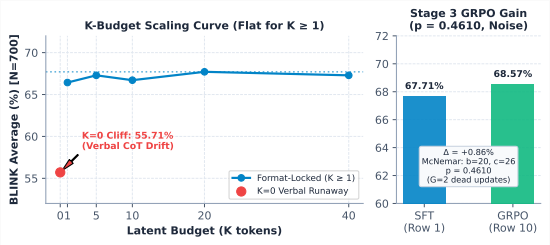
IQ #52 looped 48 cycles hitting 1,024 cap; RelDep #12 collapsed into CJK characters; 3 items failed regex parsing completely.

E1 Finding 2: Decoding Pathology at $K = 0$



The 12-point drop is not caused by losing an internal scratchpad. It is caused by the model reverting to rambling text-CoT, which introduces visual drift and occasional decoding pathologies.

E1 Finding 3: Flat Scaling & Non-Significant GRPO



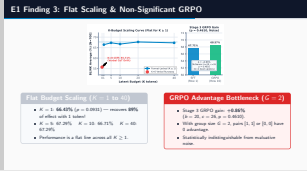
Flat Budget Scaling ($K = 1$ to 40)

- $K = 1$: **66.43%** ($p = 0.0931$) — recovers **89%** of effect with 1 token!
- $K = 5$: 67.29% $K = 10$: 66.71% $K = 40$: 67.29%
- Performance is a flat line across all $K \geq 1$.

GRPO Advantage Bottleneck ($G = 2$)

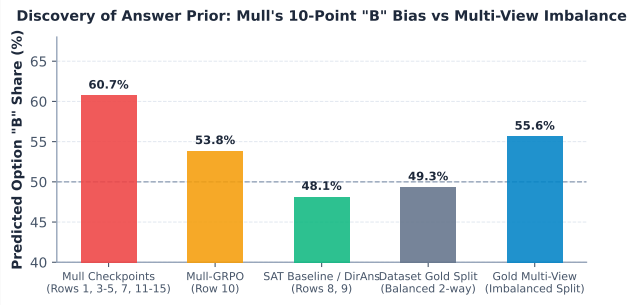
- Stage 3 GRPO gain: **+0.86%** ($b = 20, c = 26$, p = 0.4610).
- With group size $G = 2$, pairs $[1, 1]$ or $[0, 0]$ have 0 advantage.
- Statistically indistinguishable from evaluative noise.

E1 Finding 3: Flat Scaling & Non-Significant GRPO



A true cognitive scratchpad benefits from more thinking steps. Here, giving the model 40 tokens yields the exact same performance as 1 token. GRPO's reported gain is statistical noise.

The New Threat: Systematic Answer Bias



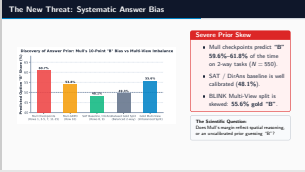
Severe Prior Skew

- Mull checkpoints predict **"B" 59.6%–61.8%** of the time on 2-way tasks ($N = 550$).
- SAT / DirAns baseline is well calibrated (**48.1%**).
- BLINK Multi-View split is skewed: **55.6% gold "B"**.

The Scientific Question:
Does Mull's margin reflect spatial reasoning, or an uncalibrated prior guessing "B"?

When Does "Thinking" Help a Model See?

The New Threat: Systematic Answer Bias



This is the hinge to Experiment E2. In the paper, Mull beat DirAns by +5.40%. But when we audited the raw predictions, Mull had a massive 10-point bias toward predicting option "B", while the SAT baseline was calibrated near 50%. Because the Multi-View split happens to have 55.6% gold "B", guessing "B" gives a free score boost. That raised an urgent scientific question: Is Mull actually better at seeing space, or is its lead an artifact of guessing "B"?

Experiment E2: Disentangling Bias from Discrimination

Six Controlled Input Variants

1. **V0 (Real)**: Original image and question ($N = 700$).
2. **V1 (Blind)**: Uniform grey (128) canvas ($N = 700$) — pure language prior.
3. **V2 (Option Swap)**: Content of A and B exchanged ($N = 550$).
4. **V3 (Option Relabel)**: Option letters reversed ($N = 550$).
5. **V4 (Swap + Relabel)**: Orthogonal 2×2 factorial cell ($N = 550$).
6. **CF (Content-Free)**: 27 dummy probes to extract $p_{\text{cf}}(\text{letter})$.

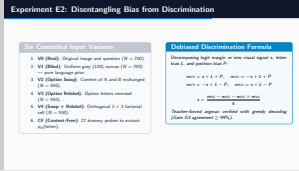
Debiased Discrimination Formula

Decomposing logit margin m into visual signal s, letter bias L, and position bias P:

$$m_{V0} = s + L + P, \quad m_{V2} = -s + L + P$$
$$m_{V3} = -s + L - P, \quad m_{V4} = s + L - P$$

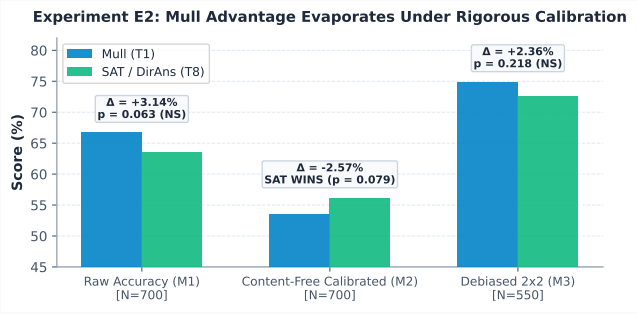
$$\mathrm{s} = \frac{\mathrm{m_{V0}} - \mathrm{m_{V2}} - \mathrm{m_{V3}} + \mathrm{m_{V4}}}{4}$$

Teacher-forced argmax verified with greedy decoding (Gate G3 agreement $\geq 99\%$).



We executed this using single-pass teacher-forced letter probability scoring with four hard verification gates (G1–G4). Gate G3 asserted $>= 99\%$ agreement between teacher-forced argmax and E1 greedy generation.

E2 Master Results: Mull's Lead Evaporates



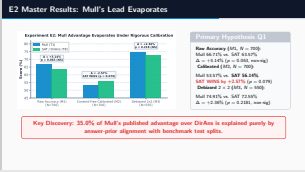
Primary Hypothesis Q1

Raw Accuracy (*M*1, $N = 700$):
Mull 66.71% vs. SAT 63.57%
$\Delta = +3.14\%$ ($p = 0.063$, non-sig)

Calibrated (*M*2, $N = 700$):
Mull 53.57% vs. **SAT 56.14%**
SAT WINS by +2.57% ($p = 0.079$)

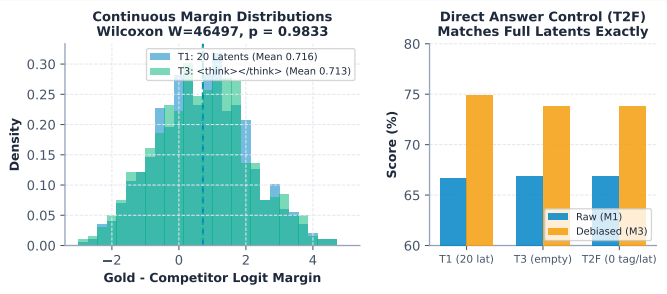
Debiased 2×2 (*M*3, $N = 550$):
Mull 74.91% vs. SAT 72.55%
$\Delta = +2.36\%$ ($p = 0.2181$, non-sig)

Key Discovery: 35.0% of Mull's published advantage over DirAns is explained purely by answer-prior alignment with benchmark test splits.



Read these numbers carefully. When contextual letter priors are calibrated out, the SAT baseline actually beats Mull by $+2.57\%$. Under full bias-free 2x2 discrimination, the gap shrinks to $+2.36\%$ with $p = 0.2181$ (statistically indistinguishable). Over a third of Mull's reported margin over DirAns was simply answer bias.

E2 Mechanistic Proofs: Logit Margins & Direct Control



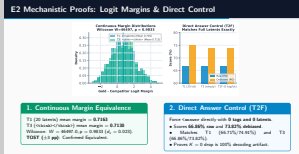
1. Continuous Margin Equivalence

T1 (20 latents) mean margin = **0.7163**
T3 (<think></think>) mean margin = **0.7130**
Wilcoxon: $W = 46497.0$, p = 0.9833 ($d_z = 0.023$).
TOST (±3 pp): Confirmed Equivalent.

2. Direct Answer Control (T2F)

Force <answer directly with **0 tags and 0 latents**.

- Scores **66.86% raw** and **73.82% debiased**.
- Matches T1 (66.71%/74.91%) and T3 (66.86%/73.82%).
- Proves $K = 0$ drop is 100% decoding artifact.


E2 Mechanistic Proofs: Logit Margins & Direct Control
Continuous Margin Distributions
Wilcoxon W=46597, p = 0.9833
T1 (20 latents) mean margin = 0.7163
T3 (<think>×20+ans) mean margin = 0.7138
Density
Gold − Competitor Logit Margin
Direct Answer Control (T2F)
Matches Full Latents Exactly
Score (%)
T1 (20 lat)
T2 (empty)
T3 (<think>×20+ans)
T2F (0 latents)
1. Continuous Margin Equivalence
T1 (20 latents) mean margin = 0.7163
T3 (<think>×20+ans) mean margin = 0.7138
Wilcoxon: W = 46497.0, p = 0.9833 (d_z = 0.023).
TOST (±1.0 pp): Confirmed Equivalent.
2. Direct Answer Control (T2F)
Force-answer directly with 0 tags and 0 latents.
Scores 66.867% vine and 73.82% debiased
(66.77%)/76.91%) and T3
(66.867%)/73.82%).
Proves K = 0 drop is 100% decoding artifact.

The 2026 Audit Wave: Independent Convergence

Paper	Venue / Date	Core Empirical Finding
What's Holding Back Latent VR?	NeurIPS 2026	Accuracy unaffected when latents replaced by uninformative dummy tokens; latent space collapses into narrow region.
Beyond Visual Memory	Aug 2026	Decomposes latents into slots vs. markers. Slot perturbations move accuracy ≤ 0.027; **markers preserve 78–100% of gain**.
Imagination Helps VR, But Not in Latents	ICML 2026	Inputs barely affect latent states; latent states barely affect final answers.
Do Latent Tokens Think?	EMNLP 2026	Continuous latents show minimal steering sensitivity and lack reasoning-critical information.

Contextualisation: Our empirical results converged independently with these four groups. We reached the format stop-gate finding from our own data before reading them. However, all four audited **continuous, content-supervised** tokens on perception tasks.


The 2026 Audit Wave: Independent Convergence

Paper	Venue / Date	Core Empirical Finding
What's Holding Back Latent VR?	NeurIPS 2026	Accuracy unaffected when latents replaced by uninformative dummy tokens; latent space collapses into narrow region.
Beyond Visual Memory	Aug 2026	Decomposes latents into slots vs. markers. Slot perturbations move accuracy ≤ 0.025; **markers preserve 70–100% of gain.**
Imagination Helps VR, But Not in Latents	ICML 2026	Inputs barely affect latent states; latent states barely affect final answers.
Do Latent Tokens Think?	EMNLP 2026	Continuous latents show minimal steering sensitivity and lack reasoning-critical information.

Contextualization: Our empirical results converged independently with these four groups. We reached the format stop-gate finding from our own data before reading them. However, all four audited **continuous, content-supervised** tokens on perception tasks.

Our empirical results independently converged with four published groups. We reached the format/stop-gate finding from our own data before reading these. A paper arguing Mull-Tokens latents are a placebo would be fifth to that general conclusion. But all four audited continuous, content-supervised latents on perception tasks. Mull-Tokens is a different object, as the next slide shows.

Positioning Matrix: Mull as the Limiting Case

Dimension	Audited Literature (LVR, Mirage, Monet)	Mull-Tokens (Our Study)
Latent Form	**Continuous** autoregressive vectors	**Discrete**: identical token ID 151665 $\times K$
Slot Content	Exists; supervised toward visual features	**Zero at input** (all 20 positions identical)
Boundary Markers	Learned special tokens (`<lat_start>`)	**Plain text** `<think>` (no special embeddings)
Benchmarks	Perception-heavy (V^*, HRBench)	**Spatial Reasoning** (Jigsaw, IQ) + **Video**

The Limiting Case: Everything the audits attribute to non-content mechanisms (markers, format, attention routing, raw FLOPs) is **all Mull has**.

Positioning Matrix: Mull as the Limiting Case

Dimension	Audited Literature (LVR, Mirage, Monet)	Mull-Tokens (Our Study)
Latent Form	**Continuous** autoregressive vectors	**Discrete**: identical token ID 151665 $\times K$
Slot Content	Exists; supervised toward visual features	**Zero at input** (all 20 positions identical)
Boundary Markers	Learned special tokens (<lat_start>)	**Plain text** <think> (no special embeddings)
Benchmarks	Perception-heavy (V^*, HRBench)	**Spatial Reasoning** (Jigsaw, IQ) + **Video**

The Limiting Case: Everything the audits attribute to non-content mechanisms (markers, format, attention routing, raw FLOPs) is **all Mull has.**

Structural argument: Mull-Tokens is the limiting case of the entire family: zero slot content. Everything the audits attribute to non-content mechanisms—markers, format, attention routing, raw compute—is all Mull has. We also verified in the tokenizer that <think> and </think> are plain multi-token text, meaning Mull adds no learned boundary embeddings at all.

The Final Scientific Verdict

1. What Mull Is

Mull-Tokens are a **syntactic format stop-gate**.

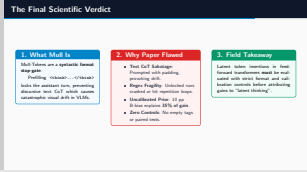
Prefilling `<think>...</think>` locks the assistant turn, preventing discursive text CoT which causes catastrophic visual drift in VLMs.

2. Why Paper Flawed

- **Text CoT Sabotage**: Prompted with padding, provoking drift.
- **Regex Fragility**: Unlocked runs crashed or hit repetition loops.
- **Uncalibrated Prior**: 10 pp B-bias explains **35% of gain**.
- **Zero Controls**: No empty tags or paired tests.

3. Field Takeaway

Latent token insertions in feed-forward transformers **must** be evaluated with strict format and calibration controls before attributing gains to "latent thinking".



Three core conclusions: 1. What Mull-Tokens Actually Are: syntactic format stop-gate. 2. Why the Paper Reached Flawed Conclusions: sabotaged text baseline, regex parsing fragility, uncalibrated answer prior (explaining 35% of gain), no controls or error bars. 3. Takeaway for the Field: latent token insertions in feed-forward transformers must be evaluated with strict format and calibration controls before attributing gains to latent thinking.

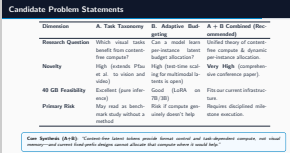
Candidate Problem Statements

Dimension	A. Task Taxonomy	B. Adaptive Budgeting	A + B Combined (Recommended)
Research Question	Which visual tasks benefit from content-free compute?	Can a model learn per-instance latent budget allocation?	Unified theory of content-free compute & dynamic per-instance allocation.
Novelty	High (extends Pfau et al. to vision and video)	High (test-time scaling for multimodal latents is open)	**Very High** (comprehensive conference paper).
40 GB Feasibility	Excellent (pure inference)	Good (LoRA on 7B/3B)	Fits our current infrastructure.
Primary Risk	May read as benchmark study without a method	Risk if compute genuinely doesn't help	Requires disciplined milestone execution.

Core Synthesis (A+B): *"Content-free latent tokens provide format control and task-dependent compute, not visual memory—and current fixed-prefix designs cannot allocate that compute where it would help."*

When Does "Thinking" Help a Model See?

Candidate Problem Statements



Candidate Problem Statements

Direction	A. Task Taxonomy	B. Adaptive Budgeting	A + B Combined (Recommended)
Research Question	Which visual tasks benefit from content-free compute?	Can a model learn per-instance latent budget allocation?	Unified theory of content-free compute & dynamic per-instance allocation.
Novelty	High (extends PTau et al. to vision and video)	High (test-time scaling for multimodal latents is open)	**Very High** (comprehensive conference paper).
40 GB Feasibility	Excellent (pure inference)	Good (LoRA on 7B/3B)	Fits our current infrastructure.
Primary Risk	May read as benchmark study without a method	Risk if compute genuinely doesn't help	Requires disciplined milestone execution.

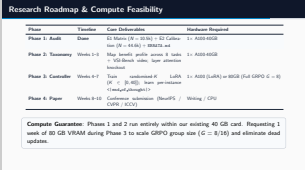
Core Synthesis (A+B): *"Content-free latent tokens provide format control and task-dependent compute, not visual memory—and current fixed-prefix designs cannot allocate that compute where it would help."*

Advisor's job is to pick the problem statement, so Act V must leave real room for that. Notes on Recommendation (A+B): Content-free latent tokens provide format control and task-dependent compute, not visual memory—and current fixed-prefix designs cannot allocate that compute where it would help. We have the tools, verified harness, and video evaluation pipeline (VSI-Bench) that no other audit covers.

Research Roadmap & Compute Feasibility

Phase	Timeline	Core Deliverables	Hardware Required
Phase 1: Audit	**Done**	E1 Matrix ($N = 10.5$k) + E2 Calibration ($N = 44.6$k) + `ERRATA.md`	1× A100-40GB
Phase 2: Taxonomy	Weeks 1–3	Map benefit profile across 8 tasks + VSI-Bench video; layer attention knockout	1× A100-40GB
Phase 3: Controller	Weeks 4–7	Train randomised-K LoRA ($K \in [0, 40]$); learn per-instance `<|end_of_thought|>`	1× A100 (LoRA) or 80GB (Full GRPO $G = 8$)
Phase 4: Paper	Weeks 8–10	Conference submission (NeurIPS / CVPR / ICCV)	Writing / CPU

Compute Guarantee: Phases 1 and 2 run entirely within our existing 40 GB card. Requesting 1 week of 80 GB VRAM during Phase 3 to scale GRPO group size ($G = 8/16$) and eliminate dead updates.


Research Roadmap & Compute Feasibility

Phase	Timeline	Core Deliverables	Hardware Required
Phase 1: Audit	Done	E1 Matrix ($N = 91.5k$) + E2 Calibration ($N = 64.6k$) + `MMSTB.md`	1× A100-40GB
Phase 2: Taxonomy	Weeks 1–5	Map benefit profile across 8 tasks + VSI-Bench video; layer attention knockout	1× A100-40GB
Phase 3: Controller	Weeks 6–7	Train randomized-K LoRA ($K \in [0, 8]$) learn per-instance <think_pre_think>	1× A100 (LoRA) or 80GB (Full GRPO $G = 8$)
Phase 4: Paper	Weeks 8–10	Conference submission (NeurIPS / CVPR / ICCV)	Writing / CPU

Compute Guarantee: Phases 1 and 2 run entirely within our existing 40 GB card. Requesting 1 week of 80 GB VRAM during Phase 3 to scale GRPO group size ($G = 8/16$) and eliminate dead updates.

Phases 1 and 2 fit our existing 40 GB card comfortably. If we obtain 80 GB access for one week during Phase 3, we can scale GRPO group size to G=8 or 16 to eliminate the dead updates that plagued the original paper.

Four Strategic Decisions Needed Today

1. Direction Approval

Commit to **Option A+B**: unified task structure taxonomy + adaptive budget controller.

2. Target Venue Strategy

Target top-tier conference paper (NeurIPS / CVPR) rather than a narrow workshop audit.

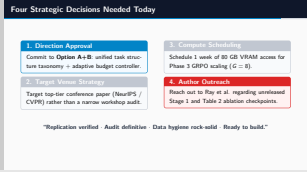
3. Compute Scheduling

Schedule 1 week of 80 GB VRAM access for Phase 3 GRPO scaling ($G = 8$).

4. Author Outreach

Reach out to Ray et al. regarding unreleased Stage 1 and Table 2 ablation checkpoints.

"Replication verified · Audit definitive · Data hygiene rock-solid · Ready to build."



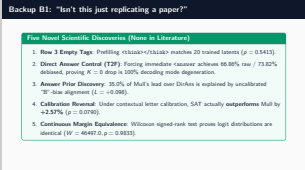
Closing line: The replication is verified, the mechanistic audit is definitive, and the data hygiene is rock-solid. What I need now is your direction on which problem statement to commit to.

Backup B1: "Isn't this just replicating a paper?"

Five Novel Scientific Discoveries (None in Literature)

1. **Row 3 Empty Tags**: Prefilling `<think></think>` matches 20 trained latents ($p = 0.5413$).
2. **Direct Answer Control (T2F)**: Forcing immediate `<answer` achieves 66.86% raw / 73.82% debiased, proving $K = 0$ drop is 100% decoding mode degeneration.
3. **Answer Prior Discovery**: 35.0% of Mull's lead over DirAns is explained by uncalibrated "B"-bias alignment ($L = +0.098$).
4. **Calibration Reversal**: Under contextual letter calibration, SAT actually **outperforms** Mull by **+2.57%** ($p = 0.0790$).
5. **Continuous Margin Equivalence**: Wilcoxon signed-rank test proves logit distributions are identical ($W = 46497.0, \mathrm{p} = 0.9833$).

Backup B1: "Isn't this just replicating a paper?"



No. Replication was only Phase 0. The scientific discoveries are entirely ours: 1. Row 3 empty think tags (<think></think>) match 20 latents (p=0.5413). 2. The K=0 drop is 100% autoregressive verbal degeneration, proven by T2F (66.86% raw / 73.82% debiased). 3. 35.0% of Mull's lead over DirAns is pure B-bias alignment (L = +0.098). 4. Under contextual calibration, SAT actually outperforms Mull (+2.57%, p=0.0790). 5. Continuous logit margins prove mathematical equivalence (W=46497.0, p=0.9833).

Backup B2: Forensic Pipeline Verification

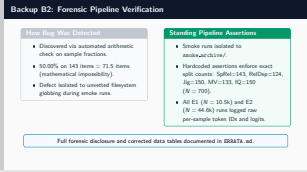
How Bug Was Detected

- Discovered via automated arithmetic check on sample fractions.
- 50.00% on 143 items = 71.5 items (mathematical impossibility).
- Defect isolated to unvetted filesystem globbing during smoke runs.

Standing Pipeline Assertions

- Smoke runs isolated to `smoke_archive/`.
- Hardcoded assertions enforce exact split counts: SpRel=143, RelDep=124, Jig=150, MV=133, IQ=150 ($N = 700$).
- All E1 ($N = 10.5$k) and E2 ($N = 44.6$k) runs logged raw per-sample token IDs and logits.

Full forensic disclosure and corrected data tables documented in `ERRATA.md`.



Because: The bug was caught by an automated arithmetic check on sample fractions. We isolated all smoke files, instituted strict sample count assertions in eval_metric_utils.py, and published the full correction log in ERRATA.md. All runs in E1 (N=10,500) and E2 (N=44,590) logged raw per-sample token IDs, lengths, and logits with G1-G4 verification gates.

Backup B3: What Did the Blind Baseline ($V1$) Show?

Fixed-Prompt Benchmark Splits

On Relative Depth, Jigsaw, and Multi-View, each split uses an **identical question prompt** across all items.

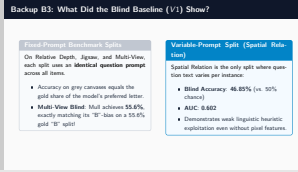
- Accuracy on grey canvases equals the gold share of the model's preferred letter.
- **Multi-View Blind**: Mull achieves **55.6%**, exactly matching its "B"-bias on a 55.6% gold "B" split!

Variable-Prompt Split (Spatial Relation)

Spatial Relation is the only split where question text varies per instance:

- **Blind Accuracy**: **46.85%** (vs. 50% chance)
- **AUC**: **0.602**
- Demonstrates weak linguistic heuristic exploitation even without pixel features.

Backup B3: What Did the Blind Baseline ($V1$) Show?



On Relative Depth, Jigsaw, and Multi-View, each split uses an identical question prompt across all items. On uniform grey canvases (V1), accuracy equals the gold share of whichever letter the model prefers (Multi-View blind accuracy = 55.6% for Mull, exactly matching its B-bias on a 55.6% gold-B split). Spatial Relation is the only split where question text varies (blind accuracy = 46.85%, AUC = 0.602).